FEATURES OF ACADEMIC ENGLISH

Academic English differs from general English in several respects. For a graduate student, being able to distinguish an academic text from other text types is as important as it is to understand that what may be acceptable in spoken language may not be appropriate in writing an essay, research paper or thesis.

Academic writing, in particular, is the product of many considerations, including but not limited to audience, purpose, organization and style. For most graduate students, the main audience for their writing is a professor, who is likely to be an expert in the subject matter. Other possible audiences include fellow students, thesis committee members and those who review their research submitted for publication or a conference presentation. Clearly, whom you are writing for will have a strong influence on the content and other aspects of your writing.

Similarly, the purpose and organization of an academic text go hand in hand. If you are writing a book review, you will follow conventions of this genre, including a general summary of the text followed by comments and evaluation of the work. On the other hand, if you are presenting original research, you will likely start your paper with an introduction followed by the methodology, results and discussion. These organizational conventions flow from the purpose of the text.

As for style, there are many descriptions of what is considered 'academic.' This diversity is complicated by the fact that some elements of academic style are discipline-specific, meaning they may differ from one area of study to another. Most would agree, however, that an academic writer´s approach to a topic can be described as objective (rather than subjective), intellectual (rather than emotional), and rational (rather than contentious). The writer's tone is serious (not conversational), impersonal (not personal), and formal (not informal).

Typical language features of academic English include:

1. Precision, clarity and sentence structures that are easy to understand, i.e.
   - No sentences that stretch for an entire paragraph (advice: read your sentences aloud to find out whether they are easily comprehensible or not.)
   - No words that prolong a sentence unnecessarily, e.g. *we analysed the data* instead of *we performed an analysis of the data.*

2. High frequency of terminology closely connected with a particular discipline, e.g. *dielectrics, perovskite solar cells, fission*

3. Sub-technical vocabulary: words and phrases common to academic writing in general, e.g. *analyse, compute, compare, define, estimate*

4. Words of Latin and Greek origin, e.g. *focus, phenomenon. data*

5. No words or phrases common in informal English, e.g. *you know, you see, a little bit, a lot of, like I said before*

6. Full forms of words instead of contracted forms, e.g. *is not* instead of *isn't*; *has not* instead of *hasn't*

7. Limited use of phrasal verbs (verb + a preposition), e.g. *constitute* instead of *make up*; *compensate* instead of *make up for*

8. Keeping verbs positive and putting negative expressions after the verb, e.g. *The analysis yielded no new results* instead of *the analysis did not yield any new results*.

9. Passive constructions instead of using "I" or "you," e.g. *the data were analysed* instead of *I analysed the data*; *a rise in consumption can be expected* instead of *you can expect a rise in consumption.*

   Notes:
   - *We* is sometimes used for variety or to emphasize that it was the author(s) who did something, e.g. *Although the method generally used for this purpose is the xx method, **we** used the method developed in our laboratory*.
   - An active construction is used where the passive construction is rather clumsy, e.g. *This paper **shows** that workshop schedules are necessary* instead of *In this paper, it **is shown***…

10. Infinitive constructions, –ing and –ed forms replacing longer phrases and clauses, e.g. *The experiment was repeated five times **to obtain** convincing results* instead of … *so that convincing results might be obtained*; ***Comparing** the two expressions we find that*… instead of *When we compare*…; *The results **obtained** do not agree with*… instead of *The results which were obtained/the results we obtained*…

11. Hedging, i.e. use of modal verbs (may, might, could, would…) and other words and phrases to avoid a definite statement, e.g.
    - The data could be interpreted as supporting industrial transformation.
    - Industrial transformation tends to be viewed as a positive development.

12. Mid-position adverbs (rather than at the initial or final position), e.g.
    - This model was ***originally*** developed by the IMF.
      (Not: This model was developed by the IMF originally.)
    - The morphology of the samples was ***then*** analysed using a microscope.
      (Not: Then the morphology of the samples was analysed using a microscope.)

Exercises:

A. Genre Comparison

1. Read the following extracts. Try to identify the type of text.
   a. a magazine for the general public
   b. a specialist magazine
   c. an advertisement
   d. a scientific paper
   e. a novel
   f. a non-fiction book of witty sayings
2. Identify features of the text that helped you make your choice.
3. Rate the degree of formality for each text (1= least formal to 4= most formal).

1.

FOR CLEANER AIR AT HOME, START WITH YOUR AIR CONDITIONER. Keep your home safe and smelling like refreshing breeze with an AC cleaner that kills 99.99% of bacteria. Get rid of all that grease, dirt, fungi, mold and other build up in your AC's cooling coils for cleaner, fresher air indoors. After cleaning, your AC will work more efficiently, and your room will feel even fresher. This is a water-based cleaner that is safe on fins and coils and doesn't need rinsing. Key Features: extends your AC's lifespan; easy to use, professional result; kills 99.9% of Staphylococcus aureus, Escherichia coli and Candida albicans.

2.

This paper presents the recognition performance obtainable, for both monochromatic and chromatic images, from Single Layer Networks (SLNs) of trixel N-Tuple (NT) nodes and SLNs of trixel Min/Max (MM) nodes. The general concepts of the trixel N-Tuple (TNT) and trixel Min/Max (TMM) techniques for the recognition of coloured images are briefly outlined. For comparative purposes, some results from implementation in 1991 are also included. From the presented results, it is concluded that, for the recognition of images with different coloured backgrounds, both the TNT and TMM networks provide greatly improved confidence levels. However, it should be noted that the documented results relate to near identical images on different coloured backgrounds and not to different images with variations in intensity, hue and saturation. Therefore, further work is required to ascertain benchmarks where the use of colour-encoded images provides a definite improvement in recognition confidence levels compared with either monochrome or single-bit binary encoded images. Also, it is essential that future systems incorporate appropriate automatic thresholding facilities where the video chain characteristics are accurately defined.

3.

Everything we do generates power – about one watt per breath, 70 watts per step. This year, Michael McAlpine and his colleagues at Princeton University figured out how to turn walking into power by embedding special crystals into a flexible, rubberlike material that, when bent, allows the crystals to produce energy. Put the crystals in shoes, say, or implant them directly into the body and they could produce enough power to charge personal electronics or internal medical devices. Elsewhere, telecommunications provider Orange introduced a prototype of Orange Power Wellies – rubber boots that convert heat into current. Campers at Britain´s Glastonbury Festival were the first to demo the footwear. (With the current model, it takes 12 hours of walking to charge a cell phone for an hour). Of course, if you assemble enough people in a tight space, they don´t even need to move to generate energy: in Paris, engineers have captured the warmth generated by bodies on the Métro (subway) to heat a public-housing project on Rue Beaubourg. By 2024, the Métro heating system will cut carbon dioxide emissions from the housing project´s heating system by a third.

4.

Murphy´s law of research
Enough research will tend to support your theory.

Smith and Holland´s law
If enough data are collected, anything may be proven by statistical methods.

Lee’s rule of accuracy
When working toward the solution of a problem, it always helps if you know the answer.

Young´s law
All great discoveries are made by mistake.

Felson´s law
To steal ideas from one person is plagiarism; to steal from many is research.

Finagle´s rule
Teamwork is essential. It allows you to blame someone else.

Meredith´s law for grad school survival
Never let your professor know that you exist.

Vile´s law for educators
No one is listening until you make a mistake.

5.

Whiz Communications Pte. Ltd. (the "Organisation") is a Singapore telecommunications service provider offering broadband internet access, local and long-distance digital IP telephony, prepaid and postpaid calling plans. On 22 April 2023, the Personal Data Protection Commission (the "Commission") was alerted by the Singapore Police Force of a personal data breach incident involving the Organisation (the "Incident"), which the Organisation confirmed on 24 April 2023. The Organisation requested, and the Commission agreed, for this matter to proceed under the Expedited Decision Breach Procedure. To this end, the Organisation voluntarily and unequivocally admitted to the facts set out in this decision and to a breach of the Protection Obligation under Section 24 of the PDPA. Section 24 of the Personal Data Protection Act 2012 ("PDPA") requires an organisation to protect personal data in its possession or control by making reasonable security arrangements to prevent unauthorised access, collection, use, disclosure, copying, modification or disposal, or similar risks (the "Protection Obligation").

6.

As research into semi-solid processing of steels has been focused on thixoforming [1], the vast potential of other new processes for creating unconventional microstructures has been neglected. Current semi-solid processing of steel typically leads to polyhedral austenite particles embedded in various forms of carbide networks [2]. Oversaturation of the solid solution with carbon provides a relatively effective stabilization of austenite [3]. Spencer et al. concluded that interventions in the evolution of such metastable microstructures may induce various transformations of this microstructure component [4]. In this paper, we will discuss several types of such interventions and the final microstructure, mechanical properties and other properties they can produce.

7.

The machine was making a strange humming sound as the door slid open. As his eyes adjusted to the light, he could see that he was not alone. There, sitting huddled in a corner of the laboratory was a small creature. "It wasn't me! I didn't do it," it squeaked at him. The apparatus was all over the place, twisted tubes and broken glass everywhere. But the machine was still working. Lights were blinking on and off, but the monitors were showing that everything was functioning normally. Apart from the small creature, there was one other thing which was odd. There was a large round hole in the floor where his desk had once been.

B. Formal and Informal Sentences

Examine the sentence pairs. Identify which one is formal (F) and informal (I). Explain your answer.

1. I showed that his arguments didn't hold water.
   It has been demonstrated that Tan's arguments are without merit.
2. I wonder why he put up with those terrible conditions for so long.
   It is not clear why such terrible conditions were tolerated for so long.
3. Five more tests will be necessary before the experiments can be concluded.
   He'll have to do a couple more tests before he can stop the experiment.
4. Please await instructions before dispatching items.
   Don´t send anything off until you´re told to do so.
5. It is possible to consider these results from a different viewpoint.
   We can think about our results in another way.
6. If we don't get a move on with our research, we'll lose our grant.
   Without expeditious progress, our research grant may be terminated.

C. Formal and Informal Paragraphs

Compare the two versions (informal and formal) of the same text and list the differences on the next page. The numbers in the brackets indicate where to look for them (e.g. the first four differences can be found in the first sentence).

Formal version	***Informal version***
1. (1, 2, 3, 4) Assembly with fasteners, such as bolts, rivets and screws, is also considered to be immediate. 2. (5, 6) However, all mechanical methods of assembly are expensive, requiring labour to drill holes and insert fasteners. 3. (7, 8, 9, 10) Manufacturers must keep an extensive inventory of fasteners on hand, and assembly with fasteners is time-consuming. 4. (11) Fasteners do not distribute load over the full area of attachment. 5. (12) Rather, they concentrate stress at the fastener site. 6. (13, 14) This concentrated stress often results in substrate failure just above the fastener hole or failure of the fastener itself. 7. (15) Both fasteners and thermal joining can cause premature joint failure and may have difficulty withstanding stresses caused by flex or vibration. 8. (16) You can test this for yourself. 9. (17) Apply shear force to a bonded joint and a bolted joint made of two strips of sheet metal. 10. (18, 19) In the bonded joint, the leading edge has slightly elevated stress than the middle, but the overall force is distributed across the entire bond area, spreading out the load. 11. (20) In fact, in many cases, this can lead to "necking" or stretching of the substrate. 12. In the bolted joint, all the force is concentrated on the bolt.	1. (1, 2, 3, 4) We usually reckon that joining stuff together with things like bolts, rivets and screws, is pretty quick. 2. (5, 6) But all mechanical methods of assembly cost loads of money, requiring labour to drill holes and insert fasteners. 3. (7, 8, 9, 10) Manufacturers have got to keep a big list of fasteners on hand, and joining stuff together with fasteners takes ages. 4. (11) Fasteners don't spread load over the full area of attachment. 5. (12) What they do is, they concentrate stress at the fastener site. 6. (13, 14) This concentrated stress often ends up in substrate failure a little bit above the fastener hole or failure of the fastener itself. 7. (15) Fasteners and thermal joining can cause joint failure and may have a tough job resisting stresses caused by flex or vibration. 8. (16) Try it out for yourself. 9. (17) Push sideways on a bonded joint and a bolted joint made of two strips of sheet metal. 10. (18, 19) In the bonded joint, the stress on the leading edge is a bit higher than in the middle, but the overall force is spread across the entire bond area, spreading out the load. 11. (20) In fact, in a lot of cases, this can lead to "necking" or stretching of the substrate. 12. In the bolted joint, all the force is concentrated on the bolt.

13. (21, 22) This concentration of stress leads to failure of the joint at approximately half the final strength of the bonded assembly. 14. (23) In addition, the holes drilled for fasteners can create leak paths – a starting point for corrosion.	13. (21, 22) This concentration of stress makes the joint fail at about half the final strength of the bonded assembly. 14. (23) The holes drilled for fasteners can create leak paths too – a starting point for corrosion.

Formal	**Informal**
Sentence 1	
(1) Assembly	joining stuff together
(2) (Assembly) is considered to be …….	……
(3) ……	……
(4) ……	……
Sentence 2	
(5) ……	……
(6) ……	……
Sentence 3	
(7) ……	……
(8) ……	……
(9) ……	……
(10) ……	……
Sentence 4	
(11) ……	……
Sentence 5	
(12) ……	……
Sentence 6	
(13) ……	……
(14) ……	……

Sentence 7

(15) …….. ……

Sentence 8

(16) …….. ……

Sentence 9

(17) …….. ……

Sentence 10

(18) …… ……

(19) …… ……

Sentence 11

(20) …….. ……

Sentence 12 (none)
Sentence 13

(21) …… ……

(22) …… ……

Sentence 14

(23) …… ……

D. Concise Sentences

This exercise gives you practice in formulating straightforward, concise and easy to understand sentences. Avoid all superfluous words.

Example:
It is possible to perform verifications of the output through a special optical multimeter.

- *It is possible to verify the output using a special optical multimeter.*
- *A special optical multimeter can be used to verify the output.*

1. We perform development in the area of electronic equipment.

   ……..

2. It is planned that this laboratory should be equipped with a thermal camera.

   ……

3. We are engaged in measuring and testing of accumulators.

   ……..

4. We perform testing activities in the field of electromagnetic compatibility.

   ……

5. Implementation of ISO9000 will ensure that all preset parameters are again reproducible, and this therefore provides a high quality and reliability.

   ……..

6. It may be difficult to make a decision regarding the appropriate method that should be used.

   ……

7. The accurate measuring of impurities is enabled by the contaminometer which the laboratory is equipped with.

   ……

8. This software in our laboratory enables us to perform designs and simulations of circuits.

   ……

9. Among the laboratory equipment are instruments enabling the measurement of electric resistance.

   ……

10. (The headline of the paragraph is: Process Management) In the area of process management we offer services and perform activities relative to modelling, analysis and optimization of industrial processes.

    ……

Adapted from:

Typical features of academic English. Retrieved 18 June 2020 from http://en.fel.zcu.cz/AE%20III%20Guidelines%20for%20Academic%20Writing/Typical%20features%20of%20academic%20English.pdf